

Case Study 5: Hotel Room Reservation

Scenario:

A hotel booking website lets customers select rooms, choose dates, and view total price before confirming.

HotelResrvation.js

```
import React, { useState } from "react";

function RoomCard(props) {

  return (

    <div style={{ border: "1px solid gray", padding: "10px", margin: "10px" }}>

      <h3>{props.room.type}</h3>

      <p>Price per Night: {"\u20B9"} {props.room.price}</p>

      <p>Available: {props.room.available ? "Yes" : "No"}</p>

      <button

        onClick={() => props.selectRoom(props.room)}

        disabled={!props.room.available}

      >

        Select Room

      </button>

    </div>

  );

}

function Summary(props) {

  return (

    <div style={{ marginTop: "20px" }}>

      <h3>Booking Summary</h3>

      <p>Check-in: {props.checkIn}</p>

      <p>Check-out: {props.checkOut}</p>

      <p>Rooms Selected: {props.selectedRooms.length}</p>

      <p>Total Price: {"\u20B9"} {props.totalPrice}</p>

      {props.selectedRooms.length > 0 ? (
```

```

        <button onClick={props.confirmBooking}>Confirm Booking</button>
      ) : (
        <p>No rooms selected yet</p>
      )}
      {props.confirmed && <h4> Reservation Confirmed!</h4>}
    </div>
  );
}

const HotelReservation = () => {
  const [checkIn, setCheckIn] = useState("");
  const [checkOut, setCheckOut] = useState("");
  const [selectedRooms, setSelectedRooms] = useState([]);
  const [confirmed, setConfirmed] = useState(false);
  const rooms = [
    { id: 1, type: "Single Room", price: 1000, available: true },
    { id: 2, type: "Double Room", price: 1800, available: true },
    { id: 3, type: "Suite", price: 3000, available: false },
  ];

  const nights = checkIn && checkOut
    ? Math.max(
      1,
      (new Date(checkOut).getTime() - new Date(checkIn).getTime()) /
        (1000 * 60 * 60 * 24)
    )
    : 0;

  const totalPrice = selectedRooms.reduce(
    (sum, room) => sum + room.price * nights,
    0
  );

  const selectRoom = (room) => {
    if (!selectedRooms.find((r) => r.id === room.id)) {
      setSelectedRooms([...selectedRooms, room]);
    }
  };
}

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    setConfirmed(false);
  }
};

const confirmBooking = () => {
  if (selectedRooms.length > 0 && checkIn && checkOut) {
    setConfirmed(true);
  }
};

return (
  <div style={{ padding: "20px" }}>
    <h2>Hotel Room Reservation</h2>
    <div style={{ marginBottom: "20px" }}>
      <label>
        Check-in: {" "}
        <input
          type="date"
          value={checkIn}
          onChange={(e) => setCheckIn(e.target.value)}
        />
      </label>
      <br />
      <label>
        Check-out: {" "}
        <input
          type="date"
          value={checkOut}
          onChange={(e) => setCheckOut(e.target.value)}
        />
      </label>
    </div>
    <div style={{ display: "flex" }}>
      {rooms.map((room) => (
        <RoomCard key={room.id} room={room} selectRoom={selectRoom} />

```

```

    ))}
  </div>

  <Summary
    checkIn={checkIn}
    checkOut={checkOut}
    selectedRooms={selectedRooms}
    totalPrice={totalPrice}
    confirmBooking={confirmBooking}
    confirmed={confirmed}
  />
</div>

);
};

export default HotelReservation;

```

App.js

```

import HotelReservation from "../components/HotelReservation";

function App() {
  return (
    <div className="App">
      <HotelReservation />
    </div>
  );
}

export default App;

```